

Task 1: Provider vs User Responsibilities

For a Zomato-style application running in the cloud:

Security Area	Cloud Provider	User
Physical data center security	✓	
Physical servers	✓	
Hypervisor/infrastructure	✓	
Application security		✓
IAM permissions		✓
User passwords		✓
Data classification		✓
Cloud network configuration	Shared	Shared
OS patching	Shared, depending on service	Shared, depending on service
Data encryption configuration	Shared	Shared
Application logs/monitoring		✓

Task 2: Create a Limited IAM User

Using AWS as the example:

1. Open AWS IAM.
2. Select Users.
3. Select Create user.
4. Enter a username, for example:
 - storage-user

5. Configure the required credentials.
6. Attach only the required storage permissions.
7. Do not grant administrator or billing permissions.
8. Create the user.
9. Verify the permissions.

Task 3: Custom Upload-Only S3 Policy

Example policy:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AllowUpload",
      "Effect": "Allow",
      "Action": [
        "s3:PutObject"
      ],
      "Resource": "arn:aws:s3:::YOUR-BUCKET-NAME/*"
    },
    {
      "Sid": "DenyDelete",
      "Effect": "Deny",
      "Action": [
        "s3:DeleteObject"
      ],
      "Resource": "arn:aws:s3:::YOUR-BUCKET-NAME/*"
    }
  ]
}
```

Replace:

- YOUR-BUCKET-NAME
- with your actual bucket name.

Task 4: Cricket Ticket Booking Roles

Create three roles/users with different permissions.

User	Role	Permissions
User 1	Booking Viewer	Read/view booking data only
User 2	Booking Editor	Read and update booking data
User 3	User Administrator	Manage user accounts

Principle

- This follows Role-Based Access Control (RBAC) and the principle of least privilege because each user receives only the permissions necessary for their job.

Task 5: Cloud Misconfiguration Incident

Example: Capital One Data Breach — 2019

A well-known example is the Capital One breach, where a misconfigured web application firewall contributed to unauthorized access to cloud-hosted data. The incident exposed data belonging to many customers.

Although this incident occurred more than three years ago, it is frequently used as a classic cloud-security case study.

Shared Responsibility Explanation

Cloud providers secure the underlying cloud infrastructure, but customers are responsible for properly configuring cloud services, applications and access controls. Applying the Shared Responsibility Model means organizations must carefully review permissions, firewall configurations, IAM policies and monitoring instead of assuming the cloud provider secures everything automatically.

If your instructor strictly requires an incident from the last 3 years, use a recent cloud-misconfiguration incident from a verified news/security source rather than using Capital One.